AP Microeconomics Curve Challenge

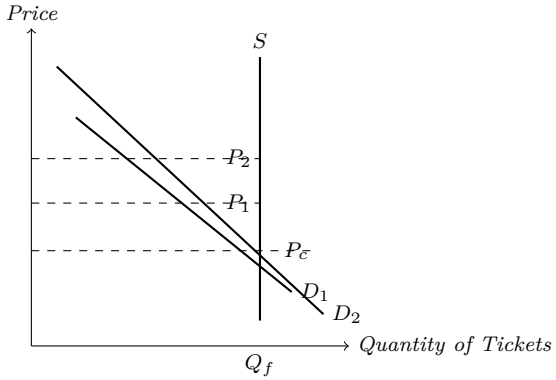
40 Hard AP-Style Graph and Curve Problems

Directions: Choose the best answer for each question. Every question in this booklet is built around a graph, curve, or curve-based inference in the style of difficult AP Microeconomics multiple-choice and FRQ graph reasoning.

Source basis: The topics, graph types, and traps in this booklet were synthesized from the 82 TB/SG/FRQ/progress-check PDFs in this folder, especially the repeated graph work on price controls, taxation, trade, perfect competition, monopoly, monopolistic competition, labor markets, monopsony, externalities, and inequality.

Graph Problems

The following questions refer to the graph below, which shows the market for tickets to an annual event with a fixed seating capacity.



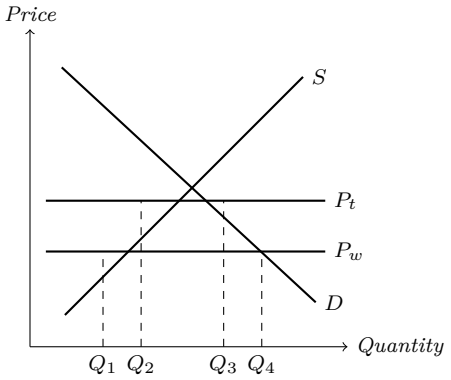
1. If demand shifts from D_1 to D_2 and no price ceiling is imposed, which of the following will occur?
   (A) Price will remain at P_1 and quantity will remain at Q_f.
   (B) Price will rise to P_2 and quantity will increase beyond Q_f.
   (C) Price will rise to P_2 and quantity will remain at Q_f.
   (D) Price will fall to P_c and quantity will remain at Q_f.
   (E) Price will rise to P_2 and quantity will fall below Q_f.

2. Suppose a binding price ceiling is set at P_c after demand has shifted to D_2. The market outcome will most likely be
   (A) quantity exchanged greater than Q_f because the lower price stimulates more demand
   (B) quantity exchanged equal to Q_f with a shortage because quantity demanded at P_c exceeds quantity supplied
   (C) quantity exchanged less than Q_f because suppliers reduce output when price falls
   (D) quantity exchanged equal to the quantity demanded on D_2 at P_c
   (E) no shortage because the original supply is vertical

3. Relative to the original equilibrium with demand curve D_1, the event organizer's total revenue after the shift to D_2 and with no price ceiling will
   (A) increase because the same number of tickets is sold at a higher price
   (B) decrease because demand becomes less elastic near the top of the graph
   (C) remain unchanged because quantity is fixed
   (D) decrease because a vertical supply curve prevents market adjustment
   (E) be indeterminate because elasticity is not labeled numerically

4. If the binding ceiling at P_c remains in place and resale is possible, which of the following is most accurate?
   (A) All buyers will necessarily pay exactly P_c because the law fixes the final transaction price.
   (B) No resale can occur because quantity supplied is perfectly inelastic.
   (C) Producer surplus must rise because the event remains sold out.
   (D) Some buyers may end up paying more than P_c in a secondary market.
   (E) The shortage disappears once buyers understand the supply curve is vertical.

The following questions refer to the graph below, which shows a small country that imports a good at world price P_w. Afterward, the country imposes a tariff that raises the domestic price to P_t.



5. Before the tariff is imposed, the quantity imported by the country is
   (A) $Q_2 - Q_1$

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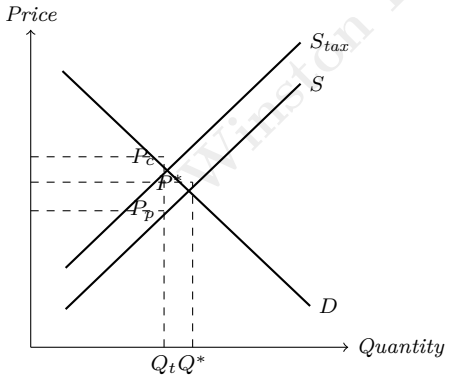
(B) $Q_3 - Q_2$
(C) $Q_4 - Q_3$
(D) $Q_4 - Q_2$
(E) $Q_4 - Q_1$

6. After the tariff is imposed and the domestic price rises to P_t, the quantity imported becomes
(A) $Q_2 - Q_1$
(B) $Q_3 - Q_2$
(C) $Q_4 - Q_3$
(D) $Q_4 - Q_1$
(E) $Q_3 - Q_1$

7. Which of the following expressions gives the government's tariff revenue?
(A) $(P_t - P_w)(Q_4 - Q_1)$
(B) $(P_t - P_w)(Q_3 - Q_2)$
(C) $(P_t - P_w)(Q_2 - Q_1)$
(D) $\frac{1}{2}(P_t - P_w)(Q_3 - Q_2)$
(E) $\frac{1}{2}(P_t - P_w)(Q_4 - Q_1)$

8. Relative to free trade, which of the following groups are guaranteed to gain from the tariff?
(A) Domestic consumers only
(B) Domestic producers only
(C) Domestic producers and the government
(D) Foreign producers and domestic consumers
(E) Domestic consumers and the government

The following questions refer to the graph below, which shows a competitive market before and after a per-unit excise tax is imposed.



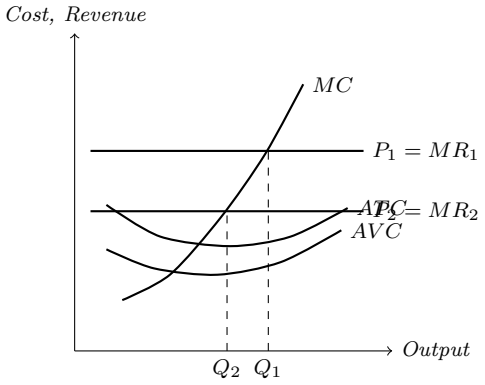
9. Based on the graph, the larger share of the tax burden is borne by
(A) consumers, because the increase from P^* to P_c is larger than the decrease from P^* to P_p
(B) producers, because the tax is legally imposed on sellers
(C) consumers and producers equally, because the tax wedge is fixed
(D) the government, because quantity falls from Q^* to Q_t
(E) consumers, but only if demand is perfectly inelastic

10. Which of the following expressions equals total tax revenue after the tax is imposed?
(A) $(P_c - P^*)Q^*$
(B) $(P^* - P_p)Q^*$
(C) $(P_c - P_p)Q_t$
(D) $\frac{1}{2}(P_c - P_p)Q_t$
(E) $(P_c - P_p)(Q^* - Q_t)$

11. The deadweight loss created by the tax is measured by
(A) $\frac{1}{2}(P_c - P_p)(Q^* - Q_t)$
(B) $(P_c - P_p)Q_t$
(C) $\frac{1}{2}(P_c - P^*)Q_t$
(D) $(P^* - P_p)(Q^* - Q_t)$
(E) $(P_c - P_p)Q^*$

12. If supply became more elastic while demand remained unchanged, which of the following would most likely occur?
(A) Consumers would bear a larger share of the tax burden.
(B) Producers would bear a larger share of the tax burden.
(C) Consumers and producers would still bear equal shares.
(D) Tax revenue would necessarily rise.
(E) Deadweight loss would necessarily fall to zero.

The following questions refer to the graph below, which shows a perfectly competitive firm's short-run cost curves and two possible market prices.



13. At price P_1, the firm's profit-maximizing output and profit situation are
(A) Q_1 and positive economic profit
(B) Q_1 and negative economic profit
(C) Q_2 and positive economic profit
(D) Q_2 and zero economic profit
(E) zero output and negative economic profit

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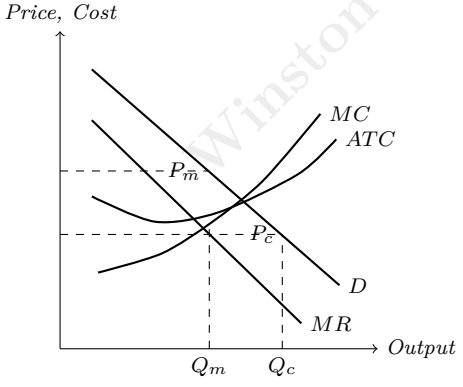
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14. At price P_2, the firm should
    (A) shut down immediately because P_2 is below average total cost
    (B) produce Q_2 because P_2 is above AVC even though the firm incurs a loss
    (C) produce Q_1 to spread fixed costs over more output
    (D) shut down because P_2 is below marginal cost at every output
    (E) produce where ATC is minimized rather than where $P_2 = MC$

15. If the firm's fixed cost increases while the market price remains at P_2, the firm's short-run profit-maximizing output will
    (A) rise above Q_2
    (B) fall below Q_2
    (C) remain at Q_2
    (D) fall to zero because ATC shifts up
    (E) become indeterminate because MC also shifts upward

16. If many identical firms in the industry are currently earning the same type of profit shown at P_1, the most likely long-run market adjustment is that
    (A) market supply will decrease and the price will rise
    (B) market supply will increase and the price will fall
    (C) market demand will decrease and the price will rise
    (D) firms will exit until price rises above P_1
    (E) the long-run equilibrium price will remain at P_1 because firms are price takers

The following questions refer to the graph below, which shows a single-price monopoly.



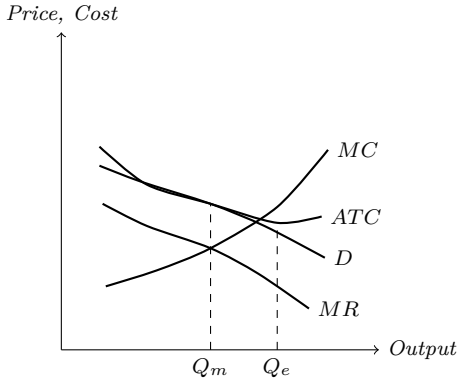
17. The monopoly's profit-maximizing output and price are
    (A) Q_c and P_c
    (B) Q_m and P_m
    (C) Q_m and P_c
    (D) Q_c and P_m
    (E) the quantity where D intersects ATC and the price at MC

18. Relative to the allocatively efficient outcome, the unregulated monopoly
    (A) produces too much because P_m exceeds ATC
    (B) produces too little because at Q_m, price exceeds marginal cost
    (C) produces the efficient quantity because $MR = MC$
    (D) produces the efficient quantity because $P_c = MC$
    (E) must be productively efficient because it is a single seller

19. If the monopolist could engage in perfect price discrimination, the quantity produced would most likely
    (A) remain at Q_m because marginal cost is unchanged
    (B) fall below Q_m because some consumers would pay more
    (C) rise to Q_c, the same quantity a perfectly competitive market would produce
    (D) rise above Q_c because all consumer surplus is captured
    (E) become indeterminate because marginal revenue is no longer defined

20. If regulators force this monopoly to charge a price equal to average total cost, the resulting output will most likely be
    (A) less than Q_m
    (B) equal to Q_m
    (C) between Q_m and Q_c
    (D) equal to Q_c
    (E) greater than Q_c

The following questions refer to the graph below, which shows a monopolistically competitive firm in long-run equilibrium.



21. The graph implies that the firm is earning
    (A) positive economic profit because the demand curve lies above marginal revenue
    (B) zero economic profit because the demand curve is tangent to the ATC curve at the profit-maximizing quantity

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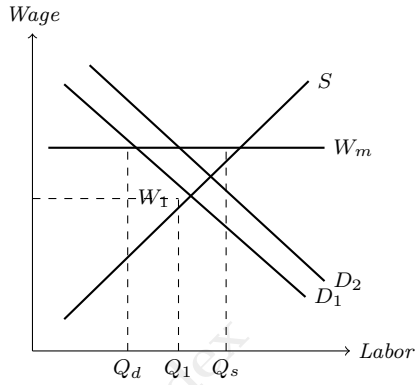
(C) negative economic profit because the MC curve is rising
(D) normal accounting profit only if price equals marginal cost
(E) a loss because output is not at the minimum point of ATC

22. The firm has excess capacity because
(A) Q_m is greater than the quantity that minimizes ATC
(B) Q_m is less than the quantity that minimizes ATC, labeled Q_e
(C) the marginal revenue curve lies below the demand curve
(D) price exceeds marginal cost
(E) its demand curve is downward sloping

23. Which of the following best describes the firm's allocative efficiency?
(A) It is allocatively efficient because demand is tangent to ATC.
(B) It is allocatively efficient because $MR = MC$ at Q_m.
(C) It is not allocatively efficient because price exceeds marginal cost at Q_m.
(D) It is not allocatively efficient because ATC exceeds AVC at all outputs.
(E) It is allocatively efficient only if the firm advertises more.

24. If the firm were instead earning positive economic profit in the short run, the long-run adjustment that would move the firm toward the graph shown is
(A) entry of new firms that shifts this firm's demand curve leftward
(B) exit of firms that shifts this firm's demand curve rightward
(C) a decrease in fixed cost that shifts ATC downward
(D) a price floor that shifts marginal revenue upward
(E) collusion among existing firms that makes demand perfectly elastic

The following questions refer to the graph below, which shows a competitive labor market. The original labor demand curve is D_1, an increase in worker productivity shifts labor demand to D_2, and the government sets a minimum wage of W_m.



25. If worker productivity increases so that labor demand shifts from D_1 to D_2, the competitive equilibrium wage and employment will
(A) both decrease
(B) both increase
(C) wage will increase and employment will decrease
(D) wage will decrease and employment will increase
(E) remain unchanged because the labor supply curve does not shift

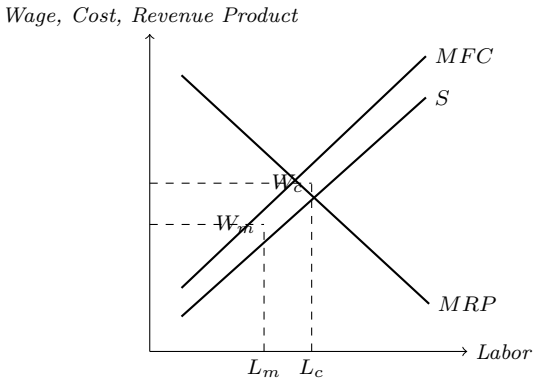
26. At the binding minimum wage W_m, employment and unemployment in this labor market are
(A) employment Q_s and unemployment $Q_s - Q_d$
(B) employment Q_d and unemployment $Q_s - Q_d$
(C) employment Q_1 and unemployment $Q_d - Q_1$
(D) employment Q_d and unemployment $Q_d - Q_s$
(E) employment Q_s and no unemployment because wage is above equilibrium

27. Total earnings received by workers who remain employed at the minimum wage are
(A) $W_1 \cdot Q_1$
(B) $W_1 \cdot Q_d$
(C) $W_m \cdot Q_d$
(D) $W_m \cdot Q_s$
(E) $W_m(Q_s - Q_d)$

28. A minimum wage like W_m is more likely to increase the total earnings of labor when labor demand is
(A) relatively elastic
(B) relatively inelastic
(C) perfectly elastic
(D) perfectly inelastic only
(E) unit elastic

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The following questions refer to the graph below, which shows a monopsonistic labor market.



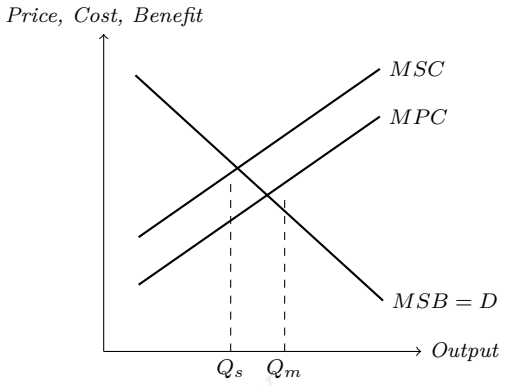
29. In the unregulated monopsony, the firm will hire
    (A) L_c workers and pay wage W_c
    (B) L_m workers and pay wage W_m
    (C) L_m workers and pay wage W_c
    (D) L_c workers and pay wage W_m
    (E) more than L_c workers because MFC exceeds S

30. Compared with a competitive labor market, the monopsony outcome results in
    (A) higher wages and higher employment
    (B) higher wages and lower employment
    (C) lower wages and lower employment
    (D) lower wages and higher employment
    (E) the same wage but lower employment

31. If the government sets a minimum wage equal to W_c, the most likely result is that employment
    (A) falls below L_m
    (B) remains at L_m
    (C) rises to L_c
    (D) rises above L_c
    (E) becomes zero because the wage is above W_m

32. The reason the MFC curve lies above the labor supply curve is that
    (A) the firm must pay a lower wage to inframarginal workers when it hires another worker
    (B) the firm can only hire labor if it also hires more capital
    (C) hiring an additional worker requires raising the wage paid to all workers
    (D) the labor supply curve is downward sloping to the firm
    (E) marginal revenue product rises faster than wage

The following questions refer to the graph below, which shows a market generating a negative production externality.



33. The unregulated market quantity is
    (A) Q_s, because the market accounts for all costs
    (B) Q_s, because MSB intersects MSC there
    (C) Q_m, because private decision makers equate marginal private cost with demand
    (D) Q_m, because marginal social cost is below marginal private cost
    (E) indeterminate because no prices are labeled

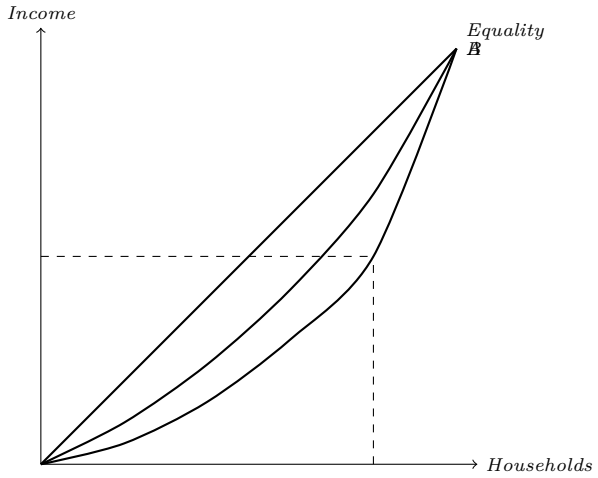
34. A per-unit Pigouvian tax that corrects the externality should be equal to
    (A) the vertical distance between MSB and MPC at Q_m
    (B) the vertical distance between MSC and MPC at Q_s
    (C) the horizontal distance between Q_m and Q_s
    (D) the vertical distance between MSB and MSC at Q_s
    (E) the entire deadweight loss triangle

35. The deadweight loss from the unregulated market is the area
    (A) between MSB and MPC from 0 to Q_s
    (B) between MSC and MPC from 0 to Q_s
    (C) between MSC and MSB from Q_s to Q_m
    (D) between MSB and MPC from Q_s to Q_m
    (E) between MSC and MPC from Q_s to Q_m plus the rectangle below it

36. If the government successfully imposes the corrective tax, which of the following will most likely occur?
    (A) Output will rise from Q_m to Q_s and total surplus will fall.
    (B) Output will fall from Q_m to Q_s and total surplus will rise.
    (C) Output will fall below Q_s and total surplus will fall.
    (D) Output will remain at Q_m and firms will simply earn lower profit.
    (E) Output will rise above Q_m because tax revenue funds public goods.

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The following questions refer to the Lorenz-curve graph below.



37. Based on the graph, which country has the more equal distribution of income?
    (A) Country A, because its Lorenz curve lies farther from the line of equality
    (B) Country A, because its top 20% receives a smaller share of income
    (C) Country B, because its Lorenz curve lies closer to the line of equality
    (D) Country B, because its Lorenz curve is steeper near the origin
    (E) The two countries are equally distributed because both end at 100%, 100%

38. Which of the following is true about the Gini coefficients of the two countries?
    (A) Country A has the smaller Gini coefficient.
    (B) Country B has the smaller Gini coefficient.
    (C) They must be equal because both countries have a Lorenz curve.
    (D) Country B's Gini coefficient must exceed 1.
    (E) The Gini coefficient cannot be related to a Lorenz curve.

39. A progressive tax-and-transfer system that reduces inequality would most likely shift a country's Lorenz curve
    (A) away from the line of equality
    (B) toward the line of equality
    (C) downward only at the 100% endpoint
    (D) into a perfectly horizontal line
    (E) into a perfectly vertical line

40. In Country A, the graph shows that the bottom 80% of households receive 50% of total income. Therefore, the top 20% of households receive
    (A) 20% of total income
    (B) 30% of total income
    (C) 40% of total income
    (D) 50% of total income
    (E) 80% of total income

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